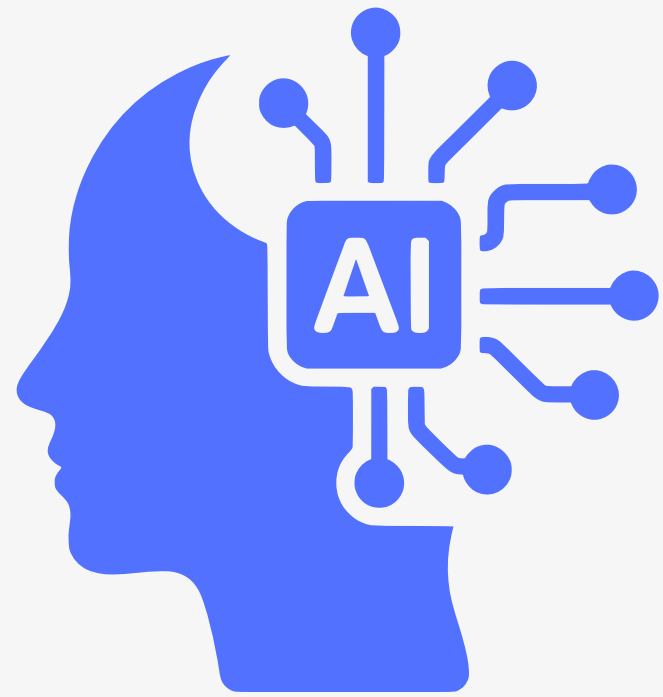




AI GUARDRAILS

IF YOUR AI DOESN'T RETURN PREDICTABLE OUTPUT, YOUR APP
WILL EVENTUALLY BREAK.





WHAT IS STRUCTURED OUTPUT?

- AI normally responds in free text.
- Free text is difficult for applications to process.
- Applications need predictable data.
- Structured output solves this problem.
- Most AI applications use structured responses.





JSON MODE

- Forces AI to return JSON only.
- Easy for applications to parse.
- Eliminates unnecessary text.
- Works well with APIs.
- Essential for backend development.
- Example: John is 25 years old and lives in London.
- ```
{ "name": "John",
```
- ```
  "age": 25,
```
- ```
 "city": "London" }
```





# PYDANTIC

- Defines the expected data structure.
- Ensures every field has the correct type.
- Automatically validates AI output.
- Prevents application crashes.
- Widely used with FastAPI and AI projects.
- Name → String
- Age → Integer
- Email → String





# VALIDATION

- Checks whether AI output is correct.
- Rejects invalid responses.
- Prevents bad data entering your system.
- Makes AI more reliable.
- Required for production systems.
- Age: ✗ Age = Twenty Five
- ☒ Age = 25







# HALLUCINATION REDUCTION

- AI sometimes invents information.
- Guardrails reduce fake responses.
- Restrict the AI's freedom.
- Ask only for supported answers.
- Validate every important response.
- If information is unavailable, return NULL.





# OUTPUT ENFORCEMENT

- Force AI to follow one format.
- No extra explanations.
- No markdown.
- No additional text.
- Only expected output.
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- No extra explanations.
- No markdown.
- No additional text.
- Only expected output.

